

Table 3: Performance of HR@N and NDCG@N on Amazon Books dataset.

Models	N=10		N=20		N=30		N=40		N=50	
	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG
BPR	0.01851	0.01710	0.02853	0.02169	0.03821	0.02564	0.04737	0.02911	0.05556	0.03205
GC-MC	0.02063	0.01898	0.03196	0.02408	0.04242	0.02835	0.05226	0.03206	0.06133	0.03532
PinSage	0.02043	0.01872	0.03210	0.02404	0.04298	0.02844	0.05239	0.03199	0.06165	0.03529
NGCF	0.02071	0.01892	0.03244	0.02425	0.04343	0.02872	0.05329	0.03243	0.06263	0.03576
<i>L-GC-MC</i>	0.02092	0.01916	0.03248	0.02443	0.04355	0.02894	0.05394	0.03286	0.06335	0.03623
<i>R-GC-MC</i>	0.01962	0.01796	0.03084	0.02307	0.04153	0.02742	0.05139	0.03115	0.06032	0.03434
<i>L-GCCF</i>	0.02067	0.01909	0.03200	0.02424	0.04312	0.02876	0.05310	0.03254	0.06218	0.03579
LR-GCCF	0.02209	0.02040	0.03407	0.02583	0.04532	0.03039	0.05532	0.03416	0.06498	0.03761

Table 4: Performance of HR@N and NDCG@N on Gowalla dataset.

Models	N=10		N=20		N=30		N=40		N=50	
	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG
BPR	0.1041	0.1011	0.1378	0.1126	0.1664	0.1221	0.1908	0.1299	0.2122	0.1365
GC-MC	0.1042	0.1010	0.1388	0.1127	0.1701	0.1222	0.1969	0.1307	0.2213	0.1381
PinSage	0.1057	0.1042	0.1390	0.1153	0.1682	0.1250	0.1935	0.1330	0.2146	0.1395
NGCF	0.1083	0.1094	0.1403	0.1197	0.1679	0.1288	0.1931	0.1368	0.2142	0.1432
<i>L-GC-MC</i>	0.1045	0.1010	0.1399	0.1132	0.1701	0.1234	0.1957	0.1316	0.2184	0.1386
<i>R-GC-MC</i>	0.1034	0.1000	0.1391	0.1123	0.1690	0.1224	0.1941	0.1305	0.2163	0.1373
<i>L-GCCF</i>	0.1044	0.1007	0.1412	0.1135	0.1721	0.1240	0.1977	0.1322	0.2196	0.1390
LR-GCCF	0.1148	0.1136	0.1518	0.1259	0.1836	0.1365	0.2113	0.1453	0.2355	0.1527

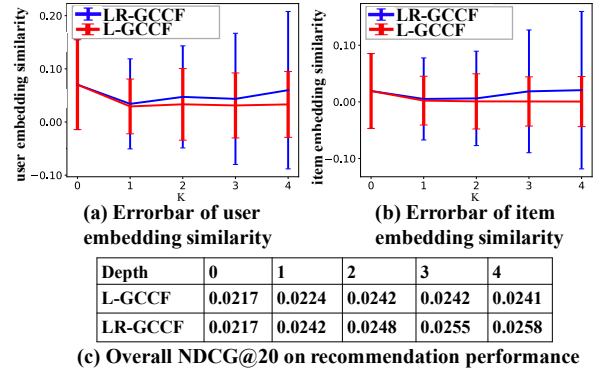
Table 5: Performance of HR@20 and NDCG@20 with different depth K.

Depth K	Amazon Books		Gowalla	
	HR@20	NDCG@20	HR@20	NDCG@20
K=0	0.0285	0.0217	0.1378	0.1126
K=1	0.0317	0.0242	0.1504	0.1246
K=2	0.0327	0.0248	0.1506	0.1248
K=3	0.0337	0.0255	0.1518	0.1259
K=4	0.0341	0.0258	0.1494	0.1241
K=5	0.0340	0.0257	0.1504	0.1247

ing could partially alleviate the over smoothing issue, and achieves better performance. Please note that, the overall trend on the Gowalla dataset is similar, and we do not show it due to page limit.

Conclusions

In this paper, we revisited the current GCN based recommendation models, and proposed a LR-GCCF model for CF based recommendation. LR-GCCF was mainly composed of two parts: First, with the recent progress of simple GCNs, we empirically removed the non-linear transformations in GCNs, and replaced it with linear embedding propagations. Second, to reduce the over smoothing effect introduced by higher layers of graph convolutions, we designed a residual preference prediction part with a residual preference learning process at each layer. Extensive experimental results clearly showed the effectiveness and efficiency of our proposed model. In the future, we would like to explore how to better integrate the representations of different layers with well defined deep neural architectures for better enhancing CF based recommendation.

Figure 3: Comparisons with and without residual preference prediction structure under different layers depth K on Amazon Books dataset.

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